1 LESSON



SCSA JavaScript Courses

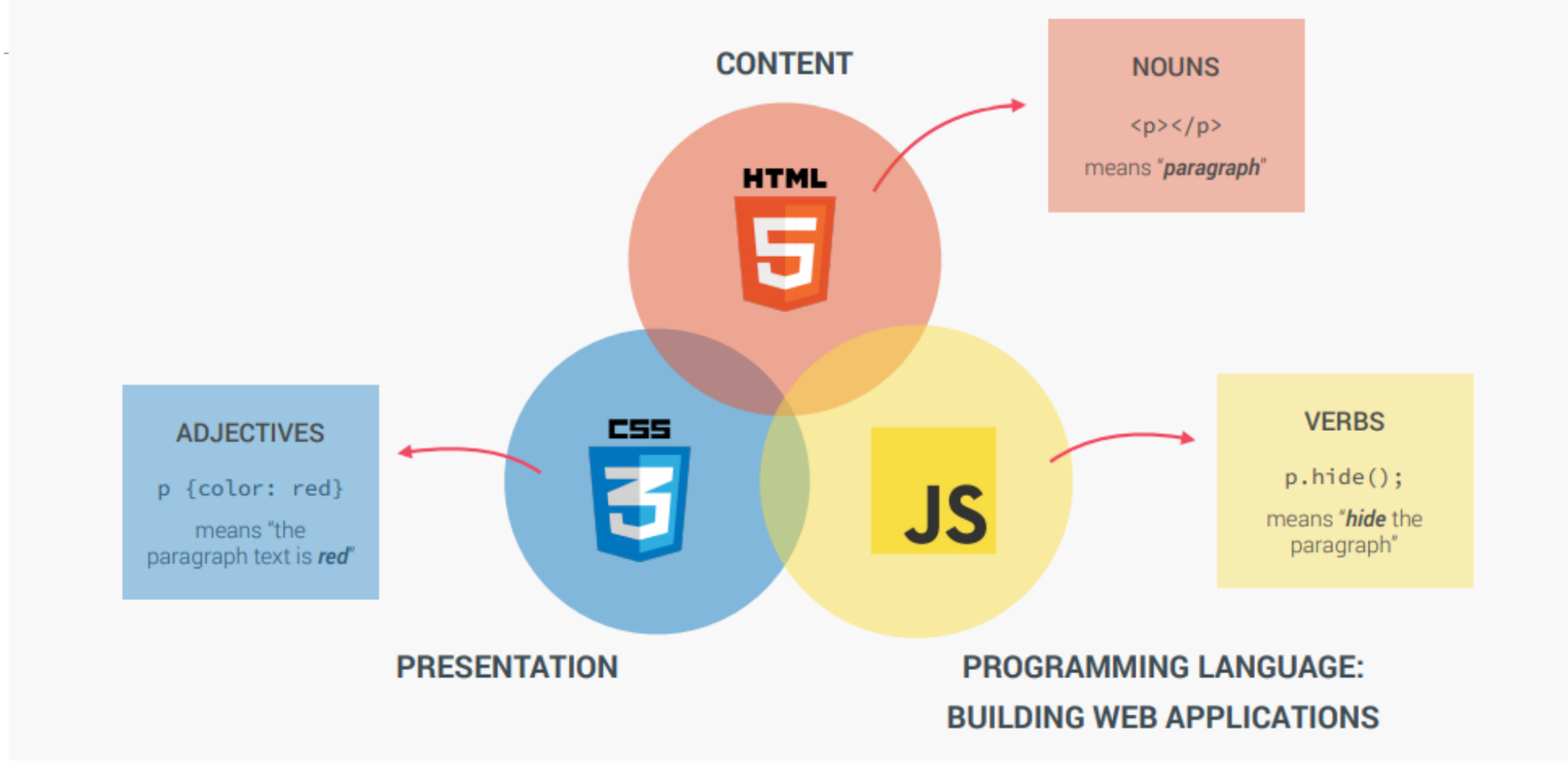
1)INTRODUCE JAVASCRIPT

2) WHAT IS LITERALS, VARIABLES AND CONSTANT

3) PRIMITIVE DATA TYPE

5) OPERATOR PRECEDENCE

Introduce Javascript



Dynamic effects and web applications in the browser

Front End apps



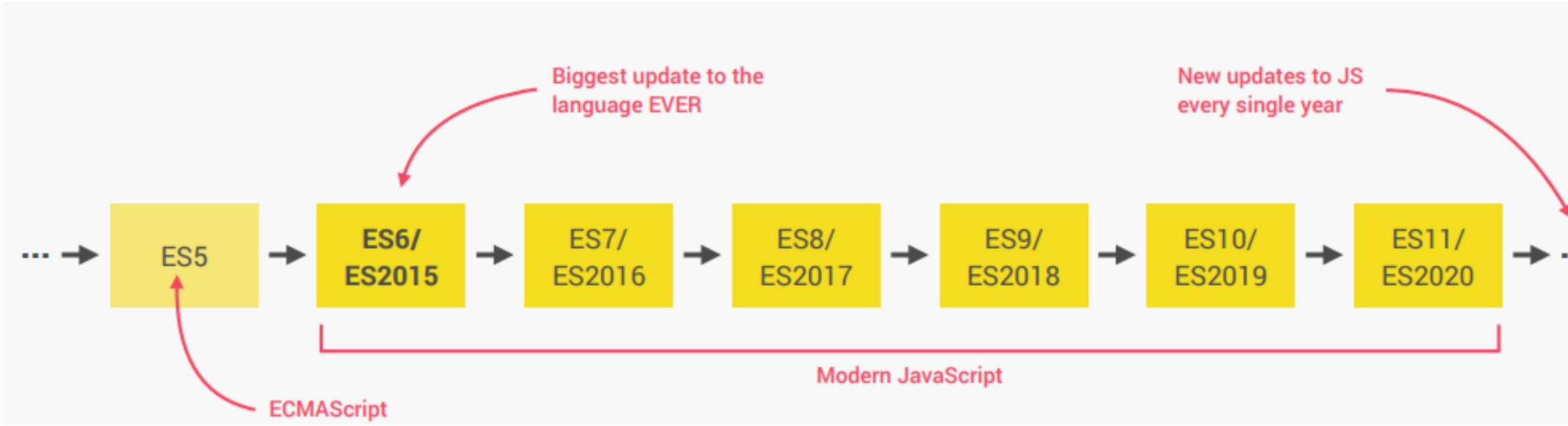


Web applications on web servers

Back-end apps





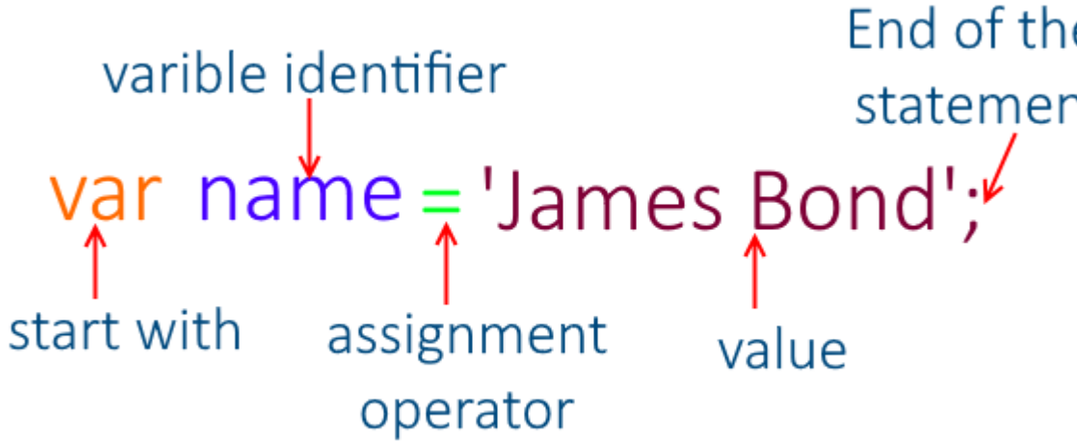

Biggest update to the language EVER
New updates to JS every single year
...
ES5
ES6/ ES2015
ES7/ ES2016
ES8/ ES2017
ES9/ ES2018
ES10/ ES2019
ES11/ ES2020
Modern JavaScript
ECMAScript

What is Literals, Variables and Constant

What is Variables?

Variable means anything that can vary. JavaScript includes variables which hold the data value and it can be changed anytime.

JavaScript uses reserved keyword **var** to declare a variable. A variable must have a unique name. You can assign a value to a variable using **equal to** (=) operator when you declare it or before us



In JavaScript variables are created using the keyword `var, let const`

In JavaScript variables are created using the keyword `var, let const`

Here you can declare the same variables more than one time and can be updated.

```
var myFriend = 'Baraka';
var myFriend = 'Peter';
console.log(myFriend); // 'Peter'
```

Let and Const

Are a new Way of declaring variables introduced in ES6 and it connot declaring twice

In JavaScript variables are created using the keyword `var, let const`
Different in var let const

```
let myFriend = 'Baraka';
let myFriend = 'Peter'; // Uncaught SyntaxError: Identifier 'myFriend' has already been declared
```

```
// let can be updated
let myFriend = 'Baraka';
myFriend = 'Peter';
console.log(myFriend); // Peter

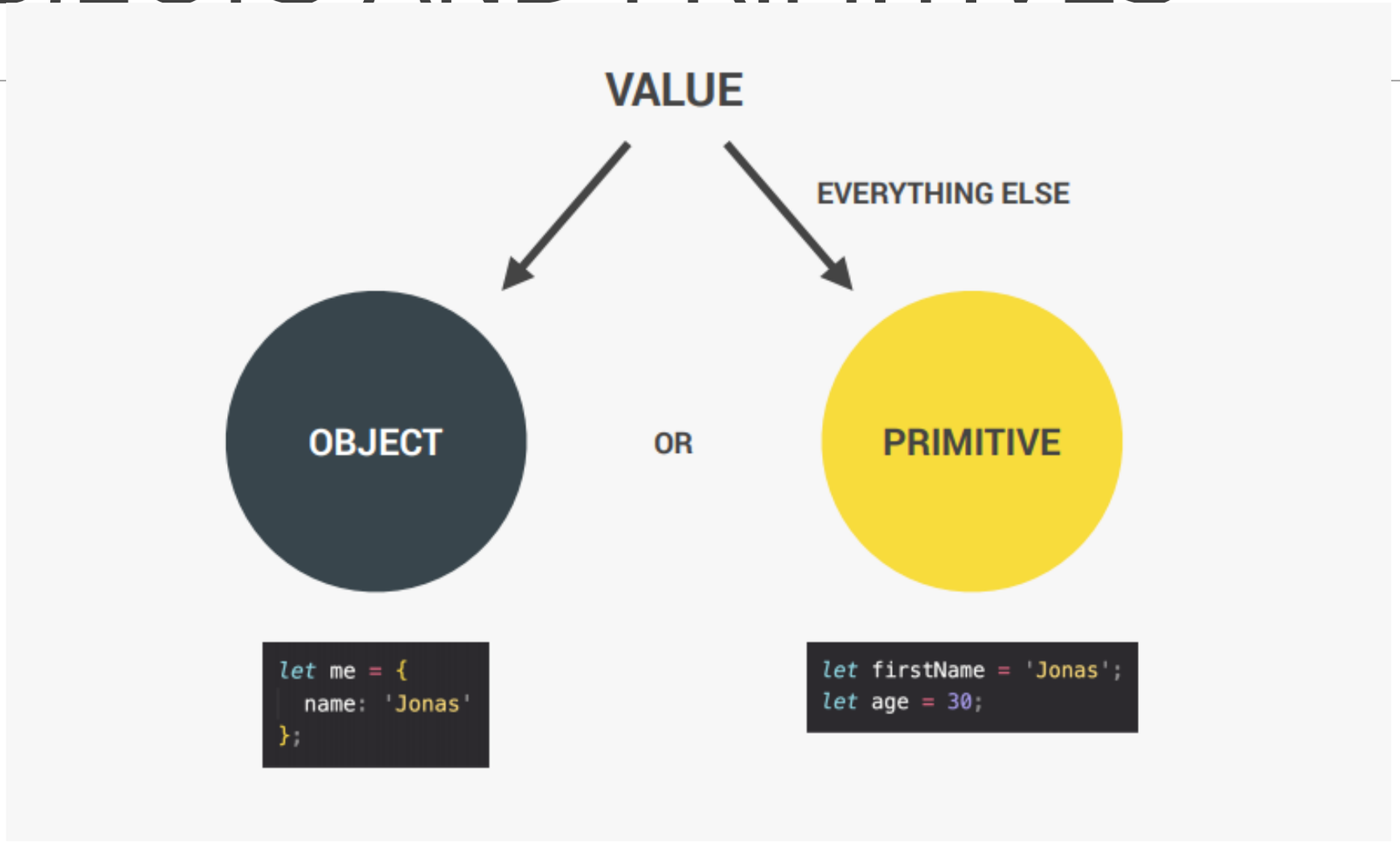
// const cannot be updated
const otherFriend = 'Patrick';
otherFriend = 'Raphael'; // Uncaught TypeError: Assignment to constant variable.
```

Also let and const are almost same, the only I known , const cannot be updated but let can

We must not use this words

abstract
boolean
break
byte
case
catch
char
class
const
continue
default
do
double
else
extends
false
final
finally
float
for
function

OBJECTS AND PRIMITIVES



Data Type

1. Number: Floating point numbers ₭ Used for decimals a `let age=50;`
2. String: Sequence of characters ₭ Used `let name="David";`
3. Boolean: Logical type that can only be true or false ₭ Used for `let fullAge=True` decisions
4. Undefined: Value taken by a variable that is not yet defined (‘empty value’) `let fullAge;`
5. Null: Also means ‘empty value’

Type Of

```
const name="daviti";
console.log(typeof(name))

const age=22;
console.log(typeof(age));

const some1=true;
console.log(typeof(some1))

let car;
console.log(typeof(car))
// reference
const hobies=['movie', 'music']
console.log(typeof(hobies))

const address={
    city:'tskhaltubo',

}
console.log(typeof(address))
```

Assigment opertors

Operator	Example	Same As
=	x = y	x = y
+=	x += y	x = x + y
-=	x -= y	x = x - y
*=	x *= y	x = x * y
/=	x /= y	x = x / y
%=	x %= y	x = x % y
**=	x **= y	x = x ** y

```
<p id="demo"></p>

<script>
txt1 = "What a very ";
txt1 += "nice day";
document.getElementById("demo").innerHTML = txt1;
</script>
```

```
<p id="demo"></p>

<script>
var a = 3;
var x = (100 + 50) * a;
document.getElementById("demo").innerHTML = x;
</script>
```

```
<script>
var x = 5 + 5;
var y = "5" + 5;
var z = "Hello" +  5;
document.getElementById("demo").innerHTML =
x + "<br>" + y + "<br>" + z;
</script>
```

String Templates

Template literals are string literals allowing embedded expressions. You can use multi-line strings and string interpolation features with them.

They were called "template strings" in prior editions of the ES2015 specification.

```
var string_name='Davit';

console.log( 'hi ' + string_name)
```

```
var string_name='Davit';

console.log( `Hi ${string_name}`)
```

template strings(backticks)
tilde

Type Conversion and Convencions

The Number() function converts the object argument to a number that represents the object's value.

If the value cannot be converted to a legal number, NaN is returned.

```
let age="18";
console.log(18 + age)
```



```
1818
```

```
let age="18";
age=Number(age)
console.log(18 + age)
```



```
36
```

```
console.log('5' - '6' - '15')
```



დავალება

გვაქვს ორი რიცხვი a, b. გამოიყვანეთ prompt ფანჯარა. სადაც ეწერება „შეიყვანე რიცხვი a-სთვის, მეორე ფანჯარა სადაც ეწერება: “შეიყვანე რიცხვი ბ-სთვის.

დავალება:

- შეამოწმე თითოეული რიცხვი მეტია თუ არა 30-ზე;
- დაუმატე ეს შეყვანილი რიცხვები ერთმანეთს;

Math Objects

Math.round()

Math.pow()

Math.sqrt()

Math.abs()

Math.ceil()

Math.floor()

Math min(), Math.max()

Math.random()

დავალება 2

გვაქვს ორი რიცხვი. X= 10, y=15;

დავალება:

- თითოეული რიცხვი აიყვანე კუბში
- შეკრიბე ორივე და ამოიღე ფესვი ამ რიცხვებიდან

საშინაო დავალება

// გამოიანგარიშე ორი მართკუთხედის ფართობი

// 1 ოთკუთხედის გვერდებია 50 და 30;

// 2 ოთკუთხედის გვერდებია 50 და 20;

// დავალება:

- გამოითვალეთ პირველის და მეორე ოთკუთხედის ფართობი
- შეამოწმე პირველი მეტია თუ არა მეორეზე;
- გამოანაგარიშე ორი ოთკუთხედის საშუალო